SAMPLE

Worlds Beyond Earth: The Formation of the Solar System

From Dust Cloud to Planets

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Introduction

Pick up a rock. Any rock: from a driveway, a riverbed, a mountaintop. Every grain of it was assembled, along with everything else you have ever touched, in an event that lasted a few tens of millions of years, four and a half billion years ago, in a cloud. The story of that event, how a shapeless knot of gas and dust became the Sun, eight planets, a hundred-plus moons, and the rock in your hand, is the story of this book, and it is one of the strangest stories science has ever worked out, because almost none of it was witnessed, and all of it had to be deduced.

The deduction took two centuries and it is still running. The idea that the Solar System condensed from a cloud is older than the discovery of galaxies; it was proposed by philosophers and mathematicians in the eighteenth century, who reasoned their way to a rotating, flattening, self-collapsing nebula decades before anyone knew what a galaxy was or how old the Earth might be. The idea then spent a century in disrepute, was revived by physics in the twentieth century, and was confirmed, in our era, by the direct observation of the very thing the philosophers imagined: young stars elsewhere in the galaxy, wearing disks of glowing dust, caught in the act of making planets. We can now watch hundreds of infant Solar Systems being born, and read the chemistry of our own in meteorites, and the story has never been better told because it has never been better known.

The narrative has a cast worth meeting in advance. There is the cloud, cold and vast, mostly hydrogen with a seasoning of every element. There is the collapse, triggered, we think, by the shock of a nearby dying star, and the disk that forms because the cloud was turning and could not stop. There is the Sun at the center, growing hot as it gathers mass, and the frost line in the disk around it, the invisible boundary where water freezes, dividing the rocky inside from the icy outside. There are the pebbles, which become boulders, which become mountain-sized planetesimals, which become planets, in a cascade that the audience will learn to picture as a series of collisions run in reverse. There are the giants, which had to finish before the gas ran out, and the migration, the discovery of the modern era, that the finished planets then drifted, stirring the whole system into the arrangement we know. And there are the leftovers, the asteroids and comets, which missed the assembly and have been keeping the receipt ever since.

The chapters follow the chronology, because the subject is an event: from the cloud to the disk, from dust to worlds, from the violent rearrangements of the youth to the quiet clockwork of the present. Along the way the book pauses on the evidence, the meteorites and the isotopes, the cratered surfaces and the sample-return missions, because the story's authority rests on them, and on the anomalies, the tilted planets and the backwards moon, the clues that keep the story unfinished.

One promise before the voyage: real science, honestly weighed, with the contested parts labeled as contested, and the wonder left in, because the wonder here is not decoration. It is the honest response to learning that the ground underfoot was manufactured by gravity, in a swirl of leftovers, in a corner of the galaxy, before there was anyone to call it a sky.

Chapter 01: An Idea Older Than the Evidence: The Nebular Hypothesis and Its History

The first people to think seriously about where the Solar System came from had almost nothing to work with. The telescopes of the eighteenth century showed planets, moons, rings, and a nebula or two, and that was the whole inventory. What the era had instead was the new mathematics of gravitation, and from that, remarkably, the correct answer was guessed.

The guessing was done, in overlapping steps, by three men whose names still label the idea.

The traveler should notice what these three did not know, because the list is the history's best measure of the achievement. They did not know the galaxies, the atom, the age of the Earth, the physics of the stars, or the existence of any other planetary system, and their case rested on two observations: the planets all circle the same way, and they all lie in nearly the same plane. The reasoning, from those two facts to a rotating, flattened, self-collapsed cloud, is a feat of disciplined imagination the field still admires, and the modern surveyor, with the disks photographed and the isotopes counted, should pay the founders the courtesy of remembering that they were right, in the essential geometry, from almost nothing. A Swedish mystic and engineer wrote of a primordial chaos condensing under gravity into a vortex of particles. A German philosopher, working in the same years, proposed that the Sun and planets condensed from a diffuse cloud, and that the system's orderly motions, all planets circling the same way in nearly the same plane, were the fossil of the cloud's rotation. And a French

mathematician, the century's greatest, gave the idea its mathematical form and its name, describing the Sun shedding rotating rings of atmosphere that condensed into planets, a mechanism wrong in detail but correct in the essential vision: the Solar System was not assembled by a builder, it condensed, as a unit, from a spinning cloud.

The nineteenth century nearly killed the idea.

The century's other fight, worth a paragraph, was about the clock. The geologists, reading the slow deposition of strata, wanted an Earth hundreds of millions of years old, and the physicists, led by the era's greatest, calculating the Sun's possible lifetime as a slowly contracting, glowing mass, capped the whole system at a few tens of millions, and the argument, conducted in print with full Victorian politeness, was won by the physicists and lost by the truth. The resolution, when radioactivity was discovered at the century's end, transformed everything at once: the Earth's rocks carried their own clocks, the Sun's engine turned out to be nuclear rather than gravitational, and the system's age opened from tens of millions to billions of years, the room the formation story had always needed and never had. The visitor should file the episode as the book's standing reminder: the formation story's numbers are not decoration, they are the argument, and the argument's history includes a century of confident error. The objection was angular momentum, and it was a real one: the Sun holds nearly all the system's mass but almost none of its spin, the planets holding nearly all the spin with a sliver of the mass, and the physicists of the era could not find a mechanism that would concentrate the rotation so unequally. A contracting cloud should have spun the Sun up, not slowed it down, and the failure led to a century of alternatives, collision hypotheses, near-miss theories, and a general sense that the nebular idea was too tidy to be true. The idea survived on intuition and the textbooks' affection, but it was a hypothesis without a mechanism.

The twentieth century re-supplied the mechanism, in pieces, as physics matured.

The disk's first photograph, the passenger should know, was not of our system. In the modern era's first years, the infrared cameras and the coronagraphs, the instruments that blot out a star's glare to see what surrounds it, caught a young nearby star wearing a belt of glowing dust, a debris disk, the first circumstellar structure ever imaged, and the picture's effect on the field was electric: the eighteenth-century drawing, the cloud wrapped in a flattened sheet, turned out to be a real object, common in the sky, and photographable. Every chapter since has been, in a sense, an argument with that first image and its thousands of successors, and the tourist should keep the timeline's shape in mind: the vision was guessed in the 1700s, the physics was built in the 1900s, and the proof arrived in the audience's own lifetime. Magnetic fields and the young Sun's wind were shown able to carry angular momentum away from the central star and into the escaping gas, answering the century's objection. The new science of fluid dynamics showed that a collapsing cloud flattens and fragments the way the philosophers had guessed, and the theory of accretion, the building of bodies by collision, supplied the assembly process the vision lacked. By the middle of the twentieth century the nebular hypothesis had returned as the nebular theory, a mechanism-backed model with quantitative teeth, and by its end the model had met evidence it never expected: photographs of disks around infant stars, the solar nebula caught in the act elsewhere, in every direction we look.

The history's lesson, worth carrying through the rest of the book, is about how the story is being told. The eighteenth century supplied the vision, the nineteenth found the flaw, the twentieth repaired the flaw and found the proof, and the twenty-first is filling in the choreography. The bystander should expect, in the chapters ahead, a mixture of settled physics and argued interpretation, because the subject is young, the evidence is meteorites and computer models, and the correct posture toward the remaining arguments is the one this book will try to hold: certainty where the evidence warrants it, and honest labels everywhere else.

One more character belongs in this history, and the reader will meet the type again whenever a hard question is settled: the stubborn skeptic. Through the middle of the twentieth century the nebular hypothesis was not the safe option but the risky one, and the people who defended it did so against a rival, the close-encounter theory, in which a passing star pulled the Sun's material into planets like taffy. The rival died hard because it explained the planets' matching orbits handsomely, and the reader should honor the argument rather than the error: the encounter theory made real predictions, including the prediction that the planets should all share the Sun's composition, and when the rocks were read, they refused. The ordinary hydrogen and helium of the Sun is missing from Earth and the other terrestrials, and a mechanism that could not lose the gas it was given failed the test. The victory went to the model that could explain an absence, which is the historian's favorite kind of lesson.

The history also keeps a founding irony the card records for its own sake. The vision was guessed in an era of gas lamps and sailing ships, argued from by a philosopher and a mathematician working decades apart, one from the shape of the disk and one from the clockwork of the orbits, and the century that followed dismissed the idea for want of evidence the century after would supply in abundance. The nebulosity photographed in the telescope age

and the disks imaged around the young stars settled, in a lifetime, a question the founders had settled only in ink. The card anchors the arc in the standard histories of the idea, entered as the opening chapter's own moral, the formation story being one of the few in science where the eighteenth century's best guess survived to be photographed whole.